SKYLARK BUSINESS INTELLIGENCE AGENT
DECISION LOG

Assignment: Skylark Drones – Business Intelligence Agent
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Purpose: Document key assumptions, technical decisions, trade-offs, and interpretation of the assignment requirements.

1. Key Assumptions

1.1 Data Source

The provided Deal Funnel and Work Order Tracker data were treated as the primary business data sources. The data was integrated through Monday.com so that the prototype could demonstrate retrieval from an operational business system rather than relying solely on static files.

1.2 Business Metrics

I interpreted the core leadership metrics as:

- **Pipeline:** Total value of currently open deals.
- **Weighted Pipeline:** Pipeline value adjusted using deal probability.
- **Billing:** Billed value relative to total work-order value.
- **Collection:** Collected value relative to the relevant billed/order value.
- **Receivable:** Outstanding amount requiring collection.
- **Risk:** Deals requiring management attention based on available business signals.
- **Data Quality:** Missing or incomplete fields that could affect business analysis.

Where source data was incomplete, the system surfaces the limitation instead of inventing values.

1.3 Interpretation of "Leadership Updates"

I interpreted leadership updates as **concise, decision-oriented business information**, rather than raw operational data.

The solution therefore focuses on answering:

- What is our current pipeline?
- Which sectors contribute most to the pipeline?
- How much has been billed and collected?
- How much remains receivable?
- Which deals require attention?
- What data-quality issues could affect decision-making?

2. Architecture & Technology Decisions

2.1 React + FastAPI

I chose **React** for the frontend and **FastAPI/Python** for the backend.

Reasoning:

- React enables rapid development of an interactive executive dashboard.
- FastAPI provides lightweight REST APIs and integrates naturally with Python-based analytics and AI components.
- Python simplifies data normalization, business calculations, and AI-agent integration.

Trade-off: A larger enterprise architecture could provide additional scalability and structure, but would have increased implementation time without materially improving the prototype within the six-hour constraint.

2.2 Monday.com API Integration

I used the **Monday.com API** for business-data retrieval.

Reasoning:

- Keeps the application connected to the operational data source.
- Allows both the dashboard and AI assistant to work from the same business data.
- Provides clear separation between data retrieval, normalization, analytics, and presentation.

Trade-off: MCP could provide a more generalized agent-tool architecture, but direct API integration allowed faster implementation and tighter control over the business-specific functionality required for the assignment.

3. Data & Analytics Decisions

3.1 Data Normalization

Raw Monday.com records are converted into normalized Deal and Work Order structures before analytics are performed.

This prevents the analytics layer and frontend from depending directly on the source-board structure and creates a consistent format for downstream calculations.

3.2 Centralized Business Logic

Business metrics such as weighted pipeline, billing rate, collection rate, receivables, sector concentration, and risk signals are calculated in the backend analytics layer.

Reasoning:

- Creates a single source of truth.
- Avoids duplicated business logic.
- Allows both the dashboard and AI assistant to use consistent calculations.

3.3 Data Quality as a Business Signal

Missing fields such as deal value, probability, sector, and collection information are surfaced through the Data Quality view.

Rather than silently filling missing values, the system makes the limitations visible because incomplete data can directly affect management decisions.

4. AI Assistant Decisions

4.1 Grounded Business AI

The AI assistant was designed as a **grounded Business Intelligence agent**, rather than a general-purpose chatbot.

It can answer questions related to:

- Pipeline
- Sector performance
- Finance
- Receivables
- Deal risks
- Data quality

The assistant uses the application's analytics/data tools to retrieve business information before generating responses.

Reasoning: Leadership-facing AI should prioritize factual and traceable business information over generating unsupported answers.

4.2 Controlled Tool-Based Approach

A controlled set of business tools was preferred over a completely open-ended analytical agent.

Trade-off: A more advanced natural-language-to-SQL system could support a wider range of analytical questions, but the controlled approach provides greater reliability and lower hallucination risk within the available development time.

5. Error Handling & Reliability

The application was designed to handle:

- Monday.com/API failures
- Backend request failures
- Missing or malformed business fields
- Empty datasets
- Invalid numerical values
- Failed AI requests

- Frontend/backend connectivity issues

The system provides fallback states and retry functionality instead of allowing failures to crash the user interface.
Missing information is represented as unavailable rather than fabricated.

6. Deployment Decision

The frontend and backend were deployed separately to satisfy the requirement for a **hosted prototype that can be tested without local setup**.

Reasoning:

- Provides direct access to the working prototype.
- Separates frontend presentation from backend/API responsibilities.
- More closely reflects a production-style web architecture.

The deployed frontend communicates with the FastAPI backend through configured API endpoints and CORS settings.

7. Key Trade-Offs

Decision	Chosen Approach	Trade-off
Frontend	React	Rapid UI development vs. larger production architecture
Backend	FastAPI/Python	Strong AI/data integration vs. additional enterprise infrastructure
Data access	Monday.com API	Direct control and implementation speed vs. less generalized than MCP
AI	Grounded tool-based agent	Higher reliability vs. less open-ended analysis
Analytics	Backend service layer	Consistent calculations vs. additional backend complexity
Data quality	Surface missing data	Greater transparency vs. incomplete metrics
Deployment	Separate frontend/backend	Clear separation vs. additional deployment configuration

8. What I Would Do Differently With More Time

With additional development time, I would prioritize:

1. **More advanced conversational analytics** — support follow-up questions, comparisons, trends, and broader business queries.
2. **Authentication and role-based access** — protect sensitive financial and operational information.
3. **Caching and performance optimization** — reduce repeated Monday.com API calls.
4. **Advanced risk scoring** — combine probability, deal age, close dates, deal value, and sales stage into a configurable score.
5. **Automated data-quality remediation** — move from identifying data gaps toward workflows for resolving them.
6. **Production observability** — add structured logging, monitoring, latency tracking, and alerts.
7. **Automated testing** — add unit and integration tests for normalization, analytics, API endpoints, and AI tools.

9. Overall Design Principle

The central design decision was to treat Skylark as a decision-support system rather than simply a dashboard or chatbot.
The dashboard provides leadership-level visibility, while the AI assistant provides conversational access to the same verified business information.
The architecture separates:
Data Retrieval → Normalization → Analytics → AI Tools → Presentation